Git2Report: Complete Self-Run & Execution Guide

Direct Commands, Step-by-Step Operator Manual & CampusThrift Reference

Target: Independent Developer Execution | **Platform:** Windows / macOS / Linux | **Status:** Production Ready

OVERVIEW & OBJECTIVE

This reference card provides **direct, copy-pasteable commands** to run Git2Report on **any Git repository** without manual intervention. Whether inspecting a local folder or a remote GitHub URL, Git2Report autonomously executes a 7-stage pipeline: extracting commits, filtering noise, clustering by architectural module, calculating verifiable code churn metrics, synthesizing AI or offline summaries, and outputting **Interactive HTML Dashboards, Markdown logs, and publication-grade PDFs**.

1. Initial Setup: Environment Activation

Open Windows PowerShell and navigate to the Git2Report installation folder, then activate the pre-configured virtual environment:

Step 1.1: Navigate to Project Directory

```
cd C:\Users\CC\OneDrive\Desktop\Project\git2report
```

Step 1.2: Activate Python Virtual Environment

```
.\.venv\Scripts\activate
```

Notice '(.venv)' will appear at the start of your terminal prompt.

2. Primary Execution Commands (Any Repository)

Select the scenario matching your target repository. Run commands directly from the activated prompt:

A. Local Repository — Full Package (HTML + PDF + Markdown) with Gemini AI

```
git2report --repo "C:\path\to\your\repo" --since 30d --llm gemini --all --out report.html
```

Generates: report.html (interactive dashboard), report.pdf (printable document), and report.md (markdown log).

B. Local Repository — Fast Offline Mode (No Internet / Zero API Required)

```
git2report --repo "C:\path\to\your\repo" --since 30d --all --out report.html
```

Constructs natural-language summaries deterministically using conventional commit messages and file diff heuristics.

C. Remote GitHub / GitLab URL (Automatic Clone & Reporting)

```
git2report --repo https://github.com/username/repository.git --since 30d --llm gemini --all --out report.html
```

Automatically clones the remote repository into cloned_repos/ and immediately triggers full report generation.

D. Adding Future Sprint Goals / Upcoming Milestones

```
git2report --repo . --since 14d --next "Finalize Payment Gateway" --next "Complete Unit Tests" --all --out report.html
```

Embeds verified future developer goals directly into the final report without AI hallucination.

3. Complete CLI Options & Parameter Reference

Customize reporting scope, target audiences, and AI behavior with these command-line flags:

Flag / Option	Allowed Values	Purpose & Default Behavior
--repo, -r	Path or URL	Path to local folder (e.g. ., ./myproject) or remote HTTPS Git URL. Default is current dir.

--since, -s	7d, 2w, 30d, YYYY-MM-DD	Lookback window. Supports days (30d), weeks (2w), hours (12h), or exact ISO dates (2025-01-01). Default is 7d.
--audience, -a	guide | client | team	**guide** = formal academic tone; **client** = user-facing features only; **team** = release notes grouped by type. Default is guide.
--llm	gemini | offline | ollama	AI summarizer engine. **gemini** uses Google Gemini; **offline** requires zero network; **ollama** uses local model. Default is offline.
--all	*boolean flag*	Writes all three formats simultaneously: .pdf, .html, and .md.
--out, -o	path/file.ext	Output file path. Extension (.html, .pdf, .md) defines default format. Default is report.pdf.
--next, -n	"text" (repeatable)	Adds developer-declared upcoming goals for next sprint/cycle.
--artifacts	folder_path/	Saves intermediate JSON and text files from all 7 pipeline steps for audit.

4. Real-World Execution Case: CampusThrift Repository

To verify production reliability, Git2Report was executed against **CampusThrift** (a full-stack university marketplace built with React, Vite, Tailwind, Node.js, Express, MongoDB, and Razorpay).

Exact CampusThrift Command Executed

```
git2report --repo cloned_repos/CampusThrift --since 2025-01-01 --llm gemini --all --out reports/campusthrift_gemini_all.html
```

Analyzed 58 commits across Frontend, Backend, and .Idea modules, generating verified reports.

Pipeline Verification Metric	Observed Value	Engineering Significance
Commits Extracted	58 total commits	Full Git lifecycle traced from repository initialization.
Noise Filtering	58 → 56 usable commits	2 empty / trivial merge commits safely filtered with audit reason.
Architectural Clustering	4 distinct modules	Categorized into *Frontend, Backend, .Idea, and Other* automatically.
Code Churn & Volume	18,835 lines modified	Strictly counted via Git tree diffs (Zero hallucination risk).
Language Breakdown	React: 78.2%, JS: 11.1%	Exact file-extension weighting calculated directly from repository.
Generated Outputs	HTML, PDF, Markdown, CSV	Saved in reports/campusthrift_gemini_all.* and reports/eval_campusthrift.csv.

5. Scientific Evaluation & Scoring Framework

Git2Report includes a built-in evaluation harness (git2report-eval) to quantitatively benchmark pipeline output against raw commit messages and unconstrained LLM baselines:

Step 5.1: Generate the Human-in-the-Loop Evaluation Sheet

```
git2report-eval sheet --repo cloned_repos/CampusThrift --since 2025-01-01 --n 10 --out reports/eval_campusthrift.csv
```

Generates a structured CSV sampling commits evenly across the timeframe with tool vs baseline sentences.

Step 5.2: Compute Comparative Accuracy & Readability Scores

```
git2report-eval score reports/eval_campusthrift.csv --out reports/eval_score.md
```

Computes 1-5 mean accuracy, readability ratings, and counts hallucinated unsupported claims.

6. Troubleshooting & Common Questions

Q: What if the terminal says 'No usable commits in that window'?
Fix: The --since parameter defaults to 7 days (7d). If the repository has had no activity recently, expand the window: --since 60d, --since 180d, or pass a starting date such as --since 2025-01-01.

Q: Where is the Google Gemini API key configured?
Fix: It is automatically read from .env in the project root. You can also pass it explicitly with --api-key YOUR_KEY or set it in PowerShell: $env:GEMINI_API_KEY="YOUR_KEY".

Q: What happens if Gemini API is unreachable or rate-limited?
Fix: Git2Report features automated fallback. If the AI call fails or produces invalid JSON, it logs a warning and automatically falls back to deterministic offline wording without crashing.

Q: Can I run this without activating the virtual environment?
Fix: Yes. Use the full path to Python: .\.venv\Scripts\python.exe -m git2report.cli --repo . --all --out report.html.